

Anomaly Detection Report

Figure 1: Unsupervised Anomaly Detection Results

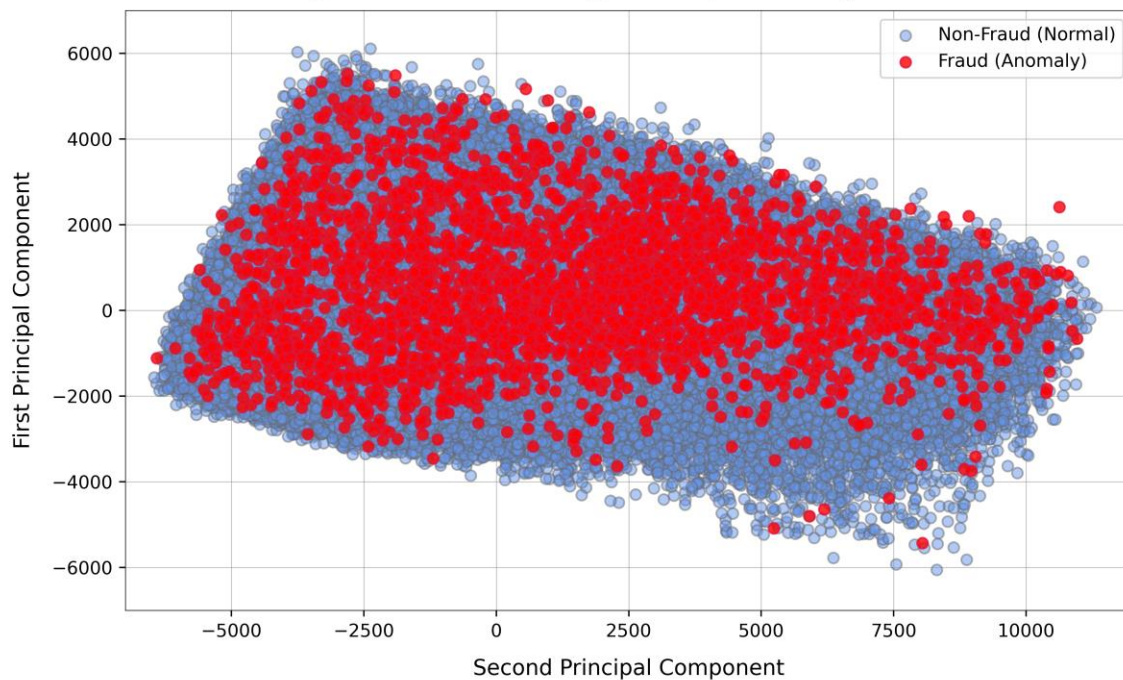
Data	True Positive Rate	AUC Score	Balanced Accuracy	F1 Score
Train	0.107	0.589	0.508	0.027
Validation	0.113	0.584	0.507	0.025
Test	0.105	0.592	0.508	0.028

Note: The dataset was partitioned into to a 60/10/30 train/validate/test split

The results above suggest that the Isolation_Forest algorithm was **unable to classify fraudulent bank account records**. This contradicts our initial hypothesis that the small proportion of fraud records (~1%) could by identified using anomaly detection.

To understand why the Isolation_Forest algorithm performed poorly, we can generate **principal components** from the test data and visualize their class labels in a scatter plot.

Figure 2: Visualizing Principal Components



The figure above shows that there is **no defined spatial separation or cluster** for records of each class. Instead, fraud and non-fraud records overlap in one big cluster.

This indicates that fraudulent bank accounts have similar feature values to regular accounts. Suggesting that a **more sophisticated model** is required to predict fraud.